

0 4

This question is about genes and the genetic code.

0 4 . 1

Define **locus**, **exon** and **universal genetic code**.

[3 marks]

Locus _____

Exon _____

Universal genetic code _____

0 4 . 2

Table 2 shows the amino acids coded for by each mRNA codon.**Table 2**

First base	Second base				Third base
	Guanine (G)	Adenine (A)	Cytosine (C)	Uracil (U)	
G	Ala	Glu	Ala	Val	G
	Gly	Glu	Ala	Val	A
	Gly	Asp	Ala	Val	C
	Gly	Asp	Ala	Val	U
A	Arg	Lys	Thr	Met	G
	Arg	Lys	Thr	Iso	A
	Ser	Asn	Thr	Iso	C
	Ser	Asn	Thr	Iso	U
C	Arg	Gln	Pro	Leu	G
	Arg	Gln	Pro	Leu	A
	Arg	Hist	Pro	Leu	C
	Arg	Hist	Pro	Leu	U
U	Trp	stop	Ser	Leu	G
	stop	stop	Ser	Leu	A
	Cyst	Tyr	Ser	Phe	C
	Cyst	Tyr	Ser	Phe	U



Do not write
outside the
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Use **Table 2** to complete **Table 3** to show:

- the mRNA base sequence transcribed from the DNA base sequence
- the amino acid sequence translated from the mRNA base sequence.

[2 marks]

Table 3

DNA base sequence	G	C	T	C	T	A	C	A	G
mRNA base sequence									
Amino acid sequence									

0 4 . 3

Deletion of the third base thymine (T) in the **DNA** base sequence shown in **Table 3** would alter the primary structure of the polypeptide produced. However, substitution of the same base would not cause any alteration.

Use information in **Table 2** to explain why.

[3 marks]

8

Turn over for the next question

Turn over ►

